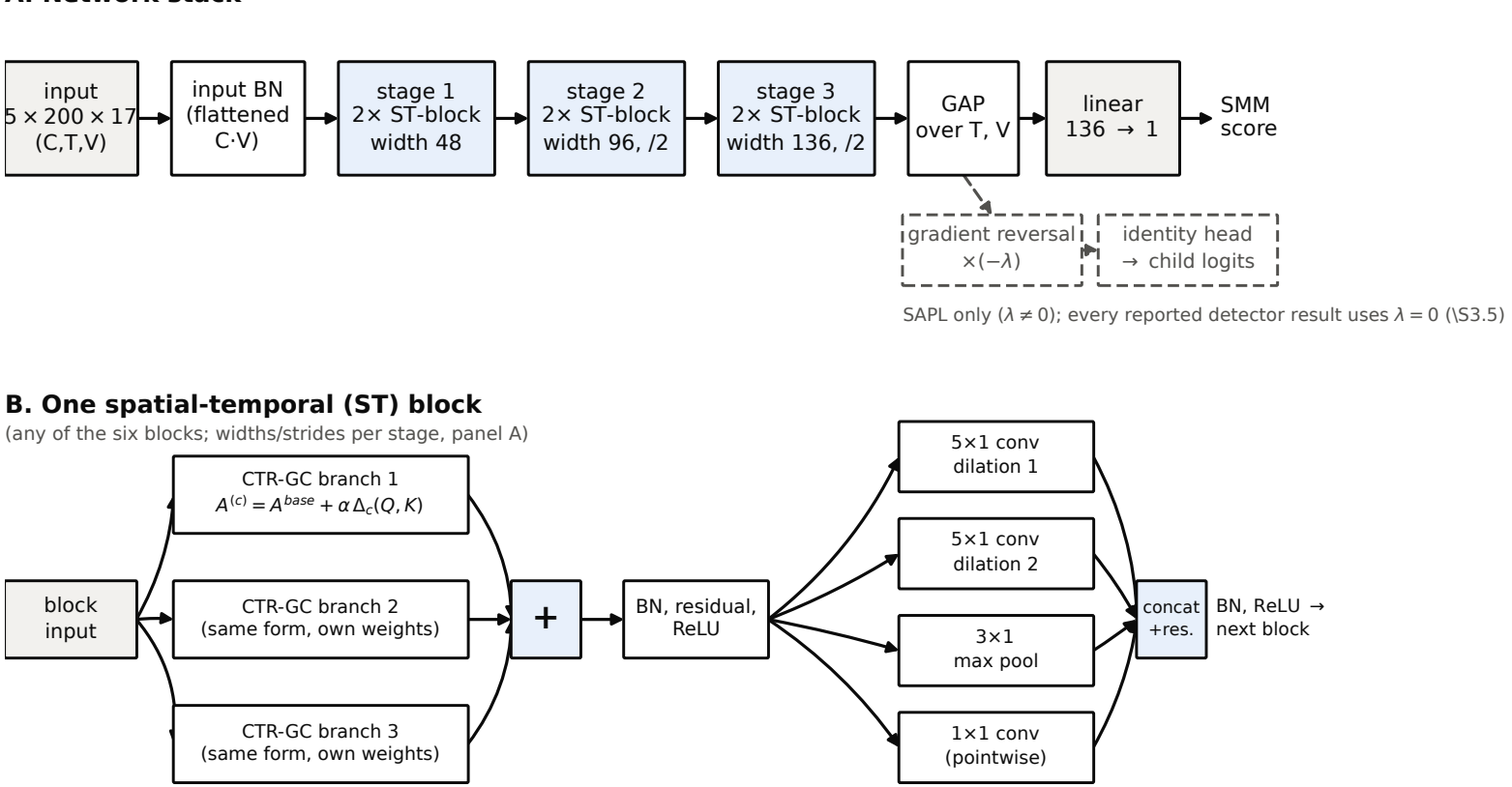
A. Network stack
input
5 × 200 × 17
(C,T,V)
input BN
(flattened
C·V)
stage 1
2× ST-block
width 48
stage 2
2× ST-block
width 96, /2
stage 3
2× ST-block
width 136, /2
GAP
over T, V
linear
136 → 1
SMM
score
gradient reversal
×(−λ)
identity head
→ child logits
SAPL only (λ ≠ 0); every reported detector result uses λ = 0 (\S3.5)
B. One spatial-temporal (ST) block
(any of the six blocks; widths/strides per stage, panel A)
block
input
CTR-GC branch 1
A(c) = Abase + α Δc(Q, K)
CTR-GC branch 2
(same form, own weights)
CTR-GC branch 3
(same form, own weights)
+
BN, residual,
ReLU
5×1 conv
dilation 1
5×1 conv
dilation 2
3×1
max pool
1×1 conv
(pointwise)
concat
+res.
BN, ReLU →
next block